Unit 3

Network Layer

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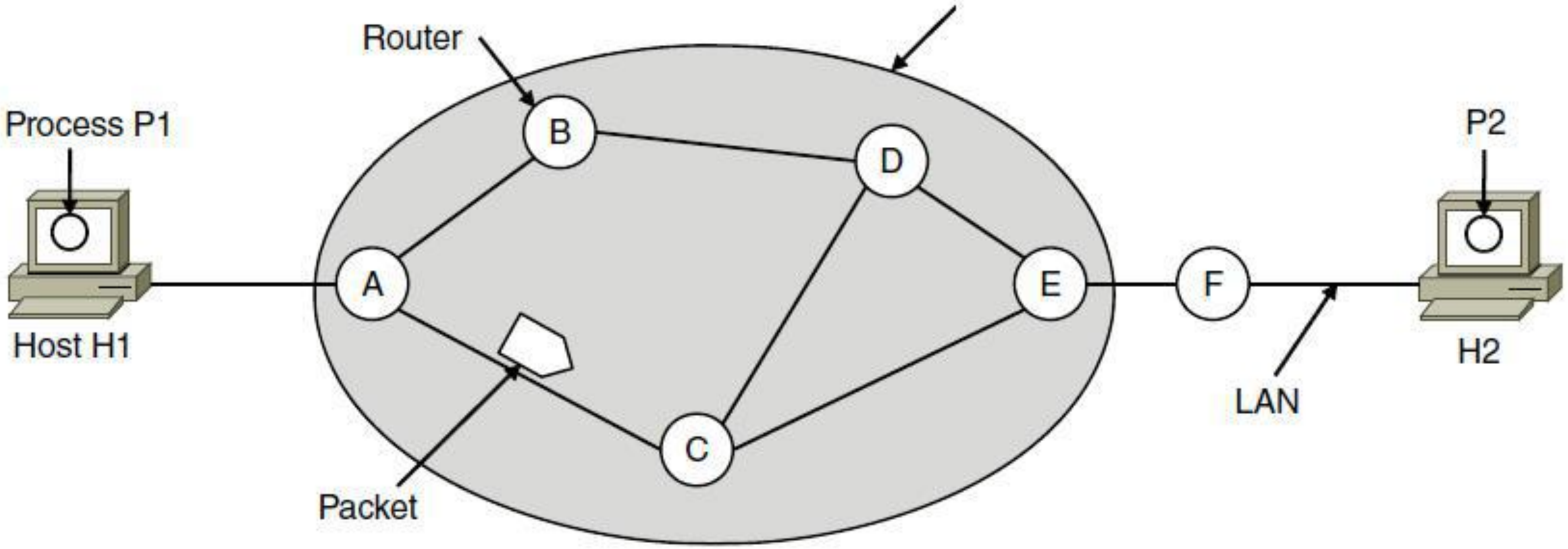
The network layer is concerned with getting packets from the source all the way to the destination. Getting to the destination may require making many hops at intermediate routers along the way. This function clearly contrasts with that of the data link layer, which has the more modest goal of just moving frames from one end of a wire to the other. Thus, the network layer is the lowest layer that deals with end-to-end transmission.

Network Layer Design Issues

1. Store-and-forward packet switching
2. Services provided to transport layer
3. Implementation of connectionless service
4. Implementation of connection-oriented service
5. Comparison of virtual-circuit and datagram networks

1. Store-and-forward packet switching

The network layer operates in an environment that uses store and forward packet switching. The node which has a packet to send, delivers it to the nearest router. The packet is stored in the router until it has fully arrived and its checksum is verified for error detection. Once, this is done, the packet is forwarded to the next router. Since, each router needs to store the entire packet before it can forward it to the next hop, the mechanism is called store − and − forward switching.



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2. Services provided to transport layer

- The network layer provides service its immediate upper layer, namely transport layer, through the network − transport layer interface. The two types of services provided are
  - Connection − Oriented Service − In this service, a path is setup between the source and the destination, and all the data packets belonging to a message are routed along this path.
  - Connectionless Service − In this service, each packet of the message is considered as an independent entity and is individually routed from the source to the destination.

The objectives of the network layer while providing these services are −

- The services should not be dependent upon the router technology.
- The router configuration details should not be of a concern to the transport layer.
- A uniform addressing plan should be made available to the transport layer, whether the network is a LAN, MAN or WAN.

3. Implementation of connectionless service

- Packet are termed as “datagrams” and corresponding subnet as “datagram subnets”. When the message size that has to be transmitted is 4 times the size of the packet, then the network layer divides into 4 packets and transmits each packet to router via. a few protocol. Each data packet has destination address and is routed independently irrespective of the packets.

4. Implementation of connection-oriented service

- To use a connection-oriented service, first we establishes a connection, use it and then release it. In connection-oriented services, the data packets are delivered to the receiver in the same order in which they have been sent by the sender.

It can be done in either two ways :

  - **Circuit Switched Connection –** A dedicated physical path or a circuit is established between the communicating nodes and then data stream is transferred.
  - **Virtual Circuit Switched Connection –** The data stream is transferred over a packet switched network, in such a way that it seems to the user that there is a dedicated path from the sender to the receiver. A virtual path is established here. While, other connections may also be using the same path.

5. Comparison of virtual-circuit and datagram networks

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

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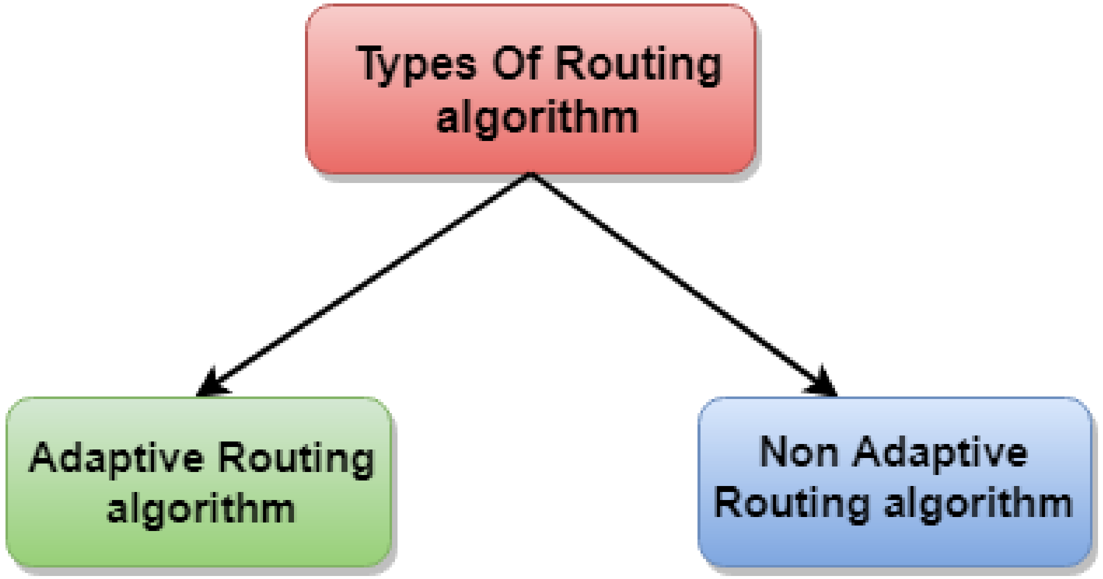
Routing Algorithms

- A routing algorithm is a procedure that lays down the route or path to transfer data packets from source to the destination. They help in directing Internet traffic efficiently. After a data packet leaves its source, it can choose among the many different paths to reach its destination. Routing algorithm mathematically computes the best path, i.e. “least – cost path” that the packet can be routed through.
- Routing algorithms can be broadly categorized into two types,

  adaptive and non adaptive routing algorithms



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- Routing algorithms can be grouped into two major classes:

  1) Non adaptive (Static Routing)

  2) Adaptive (Dynamic Routing)

Adaptive Algorithms

- Adaptive routing algorithms, also known as dynamic routing algorithms, makes routing decisions dynamically depending on the network conditions. It constructs the routing table depending upon the network traffic and topology. They try to compute the optimized route depending upon the hop count, transit time and distance.

Non – Adaptive Routing Algorithms

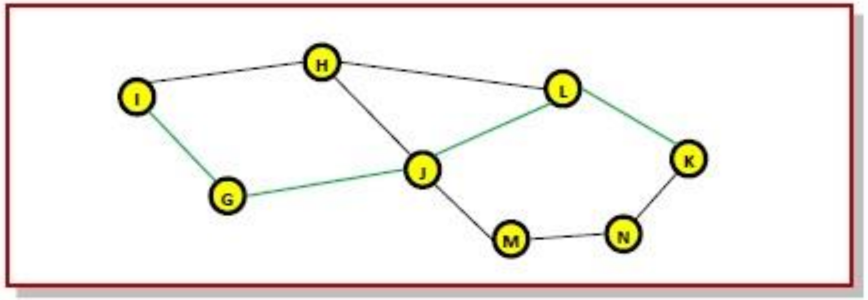
- Non-adaptive Routing algorithms, also known as static routing algorithms, construct a static routing table to determine the path through which packets are to be sent. The static routing table is constructed based upon the routing information stored in the routers when the network is booted up.

The Optimality Principle

- The purpose of a routing algorithm at a router is to decide which output line an incoming packet should go. The optimal path from a particular router to another may be the least cost path, the least distance path, the least time path, the least hops path or a combination of any of the above.
- The optimality principle can be logically proved as follows −
- If a better route could be found between router J and router K, the path from router I to router K via J would be updated via this route. Thus, the optimal path from J to K will again lie on the optimal path from I to K.

Example

- Consider a network of routers, {G, H, I, J, K, L, M, N} as shown in the figure. Let the optimal route from I to K be as shown via the green path, i.e. via the route I-G-J-L-K. According to the optimality principle, the optimal path from J to K with be along the same route, i.e. J-L-K.



- Now, suppose we find a better route from J to K is found, say along J-M-N-K. Consequently, we will also need to update the optimal route from I to K as I-G-J-M-N-K, since the previous route ceases to be optimal in this situation. This new optimal path is shown line orange lines in the following figure –
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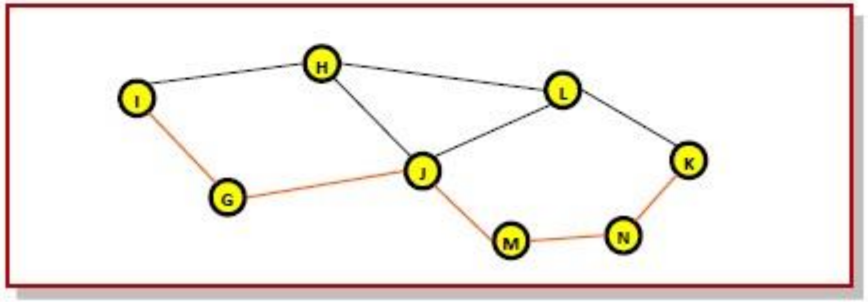

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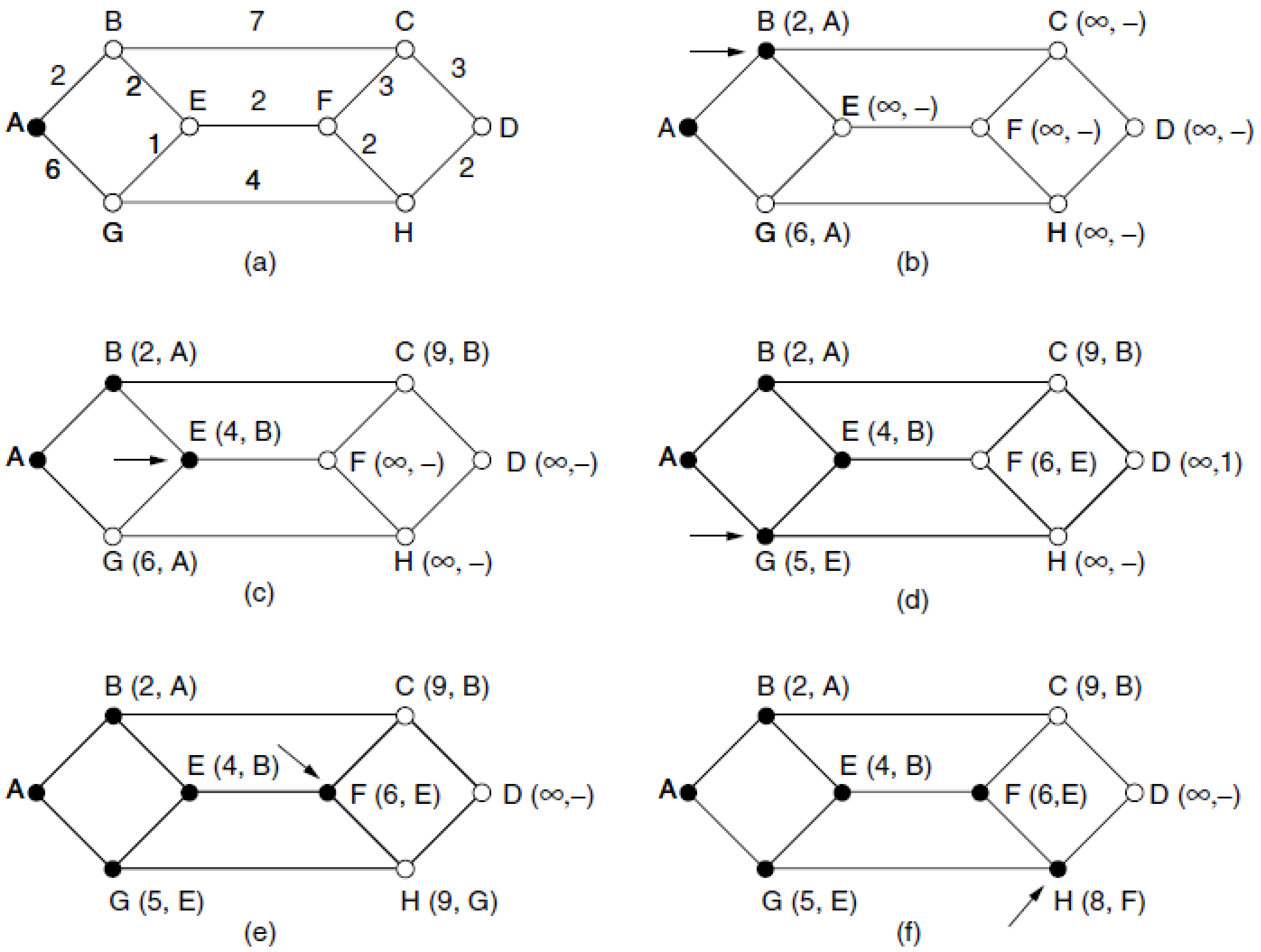
Shortest Path Algorithm

- Consider that a network comprises of N vertices (nodes or network devices) that are connected by M edges (transmission lines). Each edge is associated with a weight, representing the physical distance or the transmission delay of the transmission line. The target of shortest path algorithms is to find a route between any pair of vertices along the edges, so the sum of weights of edges is minimum. If the edges are of equal weights, the shortest path algorithm aims to find a route having minimum number of hops

Example



- The first six steps used in computing the shortest path from *A* to *D*.
- The arrows indicate the working node.

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Flooding

- When a routing algorithm is implemented, each router must make decisions based on local knowledge, not the complete picture of the network. A simple local technique is **flooding**, in which every incoming packet is sent out on every outgoing line except the one it arrived on. Flooding obviously generates vast numbers of duplicate packets, in fact, an infinite number unless some measures are taken to damp the process.
- A better technique for damming the flood is to have routers keep track of which packets have been flooded, to avoid sending them out a second time. One way to achieve this goal is to have the source router put a sequence number in each packet it receives from its hosts. Each router then needs a list per source router telling which sequence numbers originating at that source have already been seen. If an incoming packet is on the list, it is not flooded.

Distance Vector Routing (DVR)

- A **distance-vector routing (DVR)** algorithm requires that a router inform its neighbors of topology changes periodically. Historically known as the old ARPANET routing algorithm (or known as Bellman-Ford algorithm).
  - **Bellman Ford Basics –** Each router maintains a Distance Vector table containing the distance between itself and ALL possible destination nodes. Distances based on a chosen metric, are computed using information from the neighbors' distance vectors.
- Information kept by DV router - Each router has an ID
  - Associated with each link connected to a router, there is a link cost (static or dynamic).
  - Intermediate hops
- Distance Vector Table Initialization –
  - Distance to itself = 0
  - Distance to ALL other routers = infinity number.

Algorithm –

- A router transmits its distance vector to each of its neighbors in a routing packet.
- Each router receives and saves the most recently received distance vector from each of its neighbors.
- A router recalculates its distance vector when:
  - It receives a distance vector from a neighbor containing different information than before.
  - It discovers that a link to a neighbor has gone down.
- The DV calculation is based on minimizing the cost to each destination
  - **Note**
    - From time-to-time, each node sends its own distance vector estimate to neighbors.
    - When a node x receives new DV estimate from any neighbor v, it saves v's distance vector and it updates its own DV using B-F equation:

$$D_x(y) = \min \{ C(x,v) + D_v(y), D_x(y) \} \text{ for each node } y \in N$$

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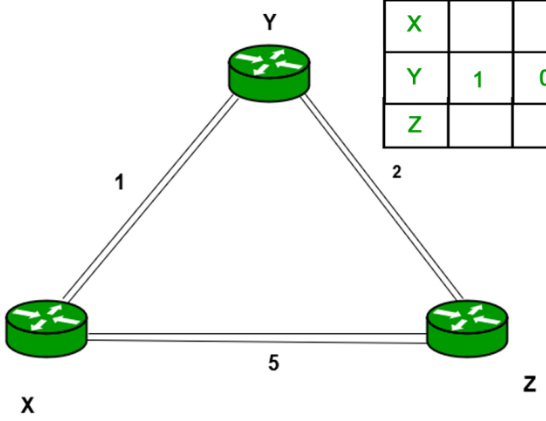
Example – Consider 3-routers X, Y and Z as shown in figure. Each router have their routing table. Every routing table will contain distance to the destination nodes.



	X	Y	Z
X	0	1	5
Y			
Z			

	X	Y	Z
X			
Y	1	0	2
Z			

	X	Y	Z
X			
Y			
Z	5	2	0

- Consider router X , X will share it routing table to neighbors and neighbors will share it routing table to it to X and distance from node X to destination will be calculated using bellmen- ford equation.
- As we can see that distance will be less going from X to Z when Y is intermediate node(hop) so it will be update in routing table X.

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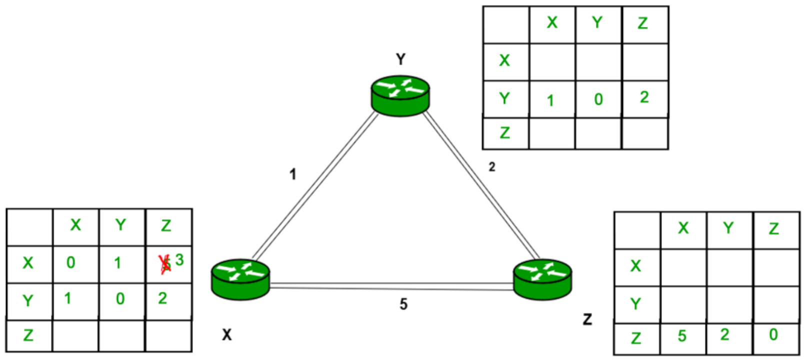
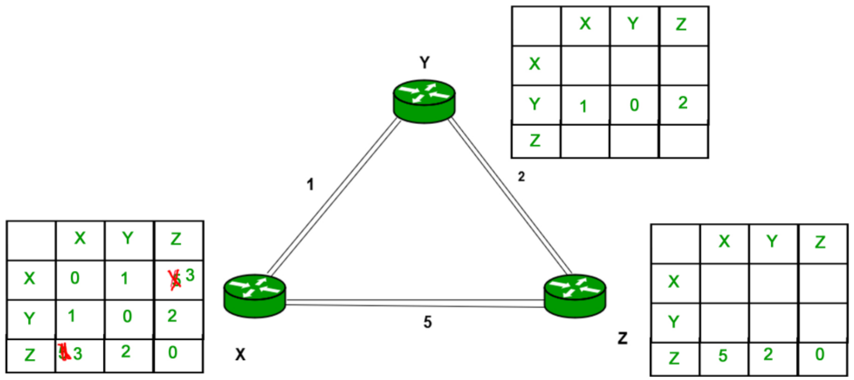
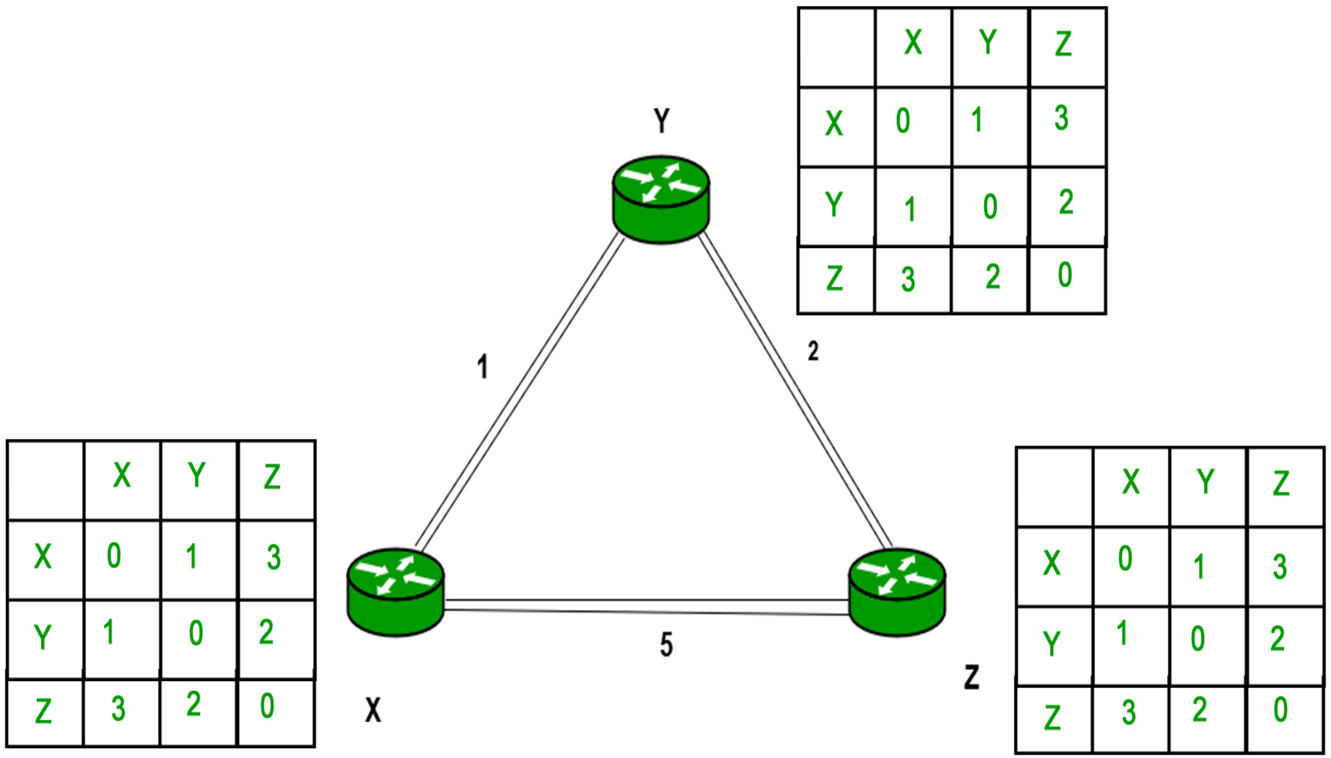


Similarly for Z also –



Finally the routing table for all –



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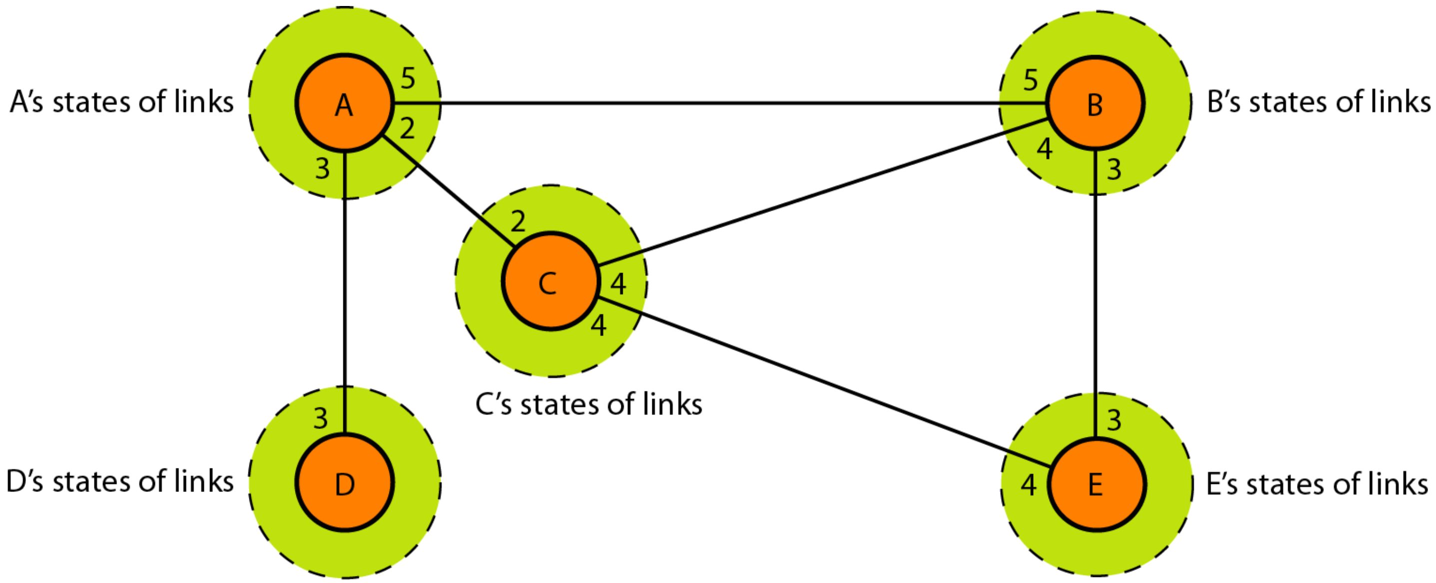
Link State Routing

- Link state routing is based on the assumption that, although the global knowledge about the topology is not clear, each node has partial knowledge: it knows the state (type, condition, and cost) of its links. In other words, the whole topology can be compiled from the partial knowledge of each node



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Building Routing Tables

1. Creation of the states of the links by each node, called the link state packet (LSP).
2. Dissemination of LSPs to every other router, called flooding, is an efficient and reliable way.
3. Formation of a shortest path tree for each node.
4. Calculation of a routing table based on the shortest path tree.

1. Creation of Link State Packet (LSP)

- A link state packet can carry a large amount of information. For the moment, we assume that it carries a minimum amount of data: the node identity, the list of links, a sequence number, and age.
- The first two, node identity and the list of links, are needed to make the topology.
- The third, sequence number, facilitates flooding and distinguishes new LSPs from old ones. The fourth, age, prevents old LSPs from remaining in the domain for a long time.

LSPs are generated on two occasions:

i. When there is a change in the topology of the domain

ii. on a periodic basis: The period in this case is much longer compared to distance vector.

- The timer set for periodic dissemination is normally in the range of 60 min or 2 h based on the implementation. A longer period ensures that flooding does not create too much traffic on the network.

2. Flooding of LSPs:

- After a node has prepared an LSP, it must be disseminated to all other nodes, not only to its neighbors. The process is called flooding and based on the following
- i. The creating node sends a copy of the LSP out of each interface
- ii. A node that receives an LSP compares it with the copy it may already have. If the newly arrived LSP is older than the one it has (found by checking the sequence number), it discards the LSP. If it is newer, the node does the following:
- a. It discards the old LSP and keeps the new one.
- b. It sends a copy of it out of each interface except the one from which the packet arrived. This guarantees that flooding stops somewhere in the domain (where a node has only one interface).

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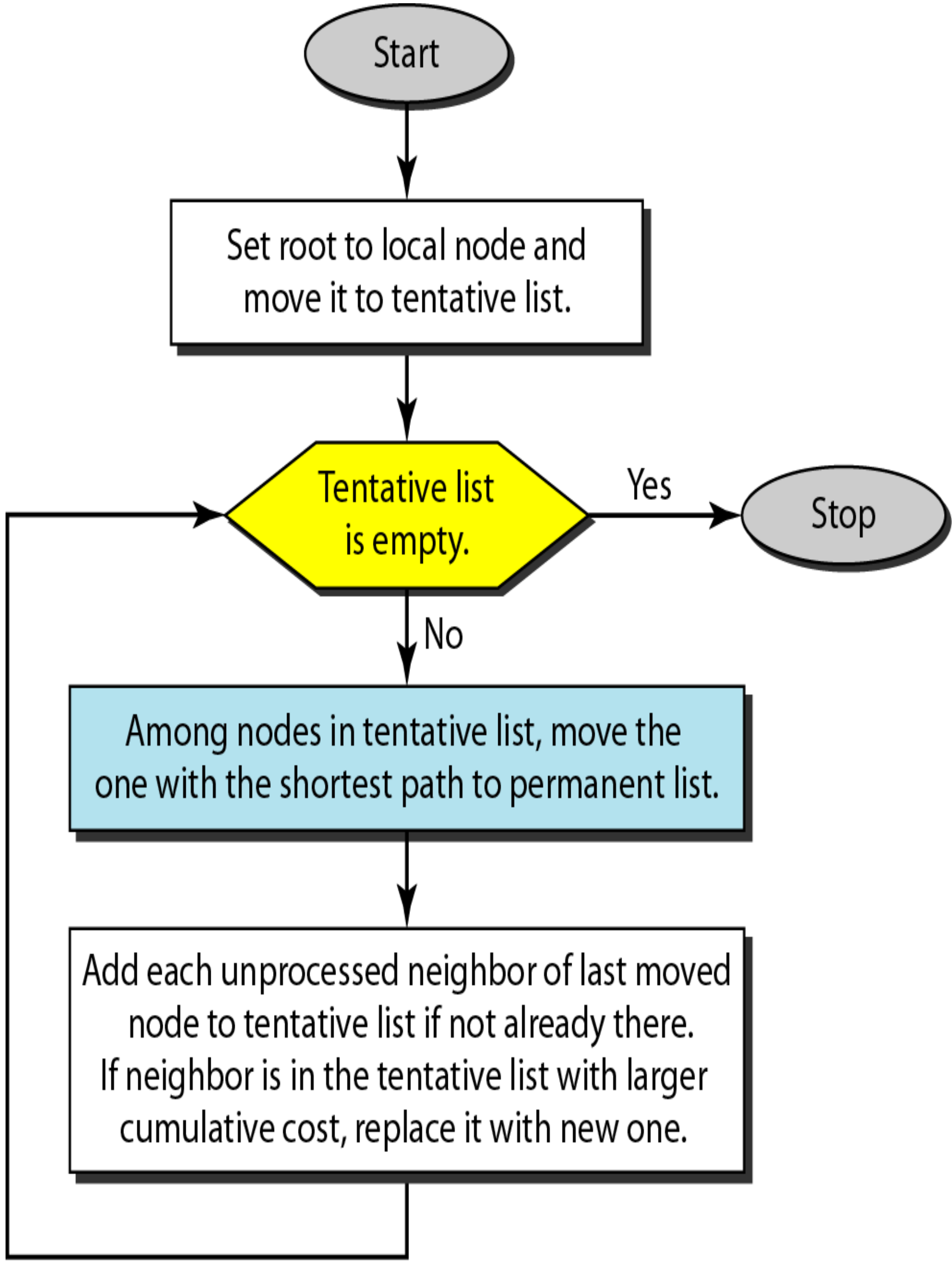
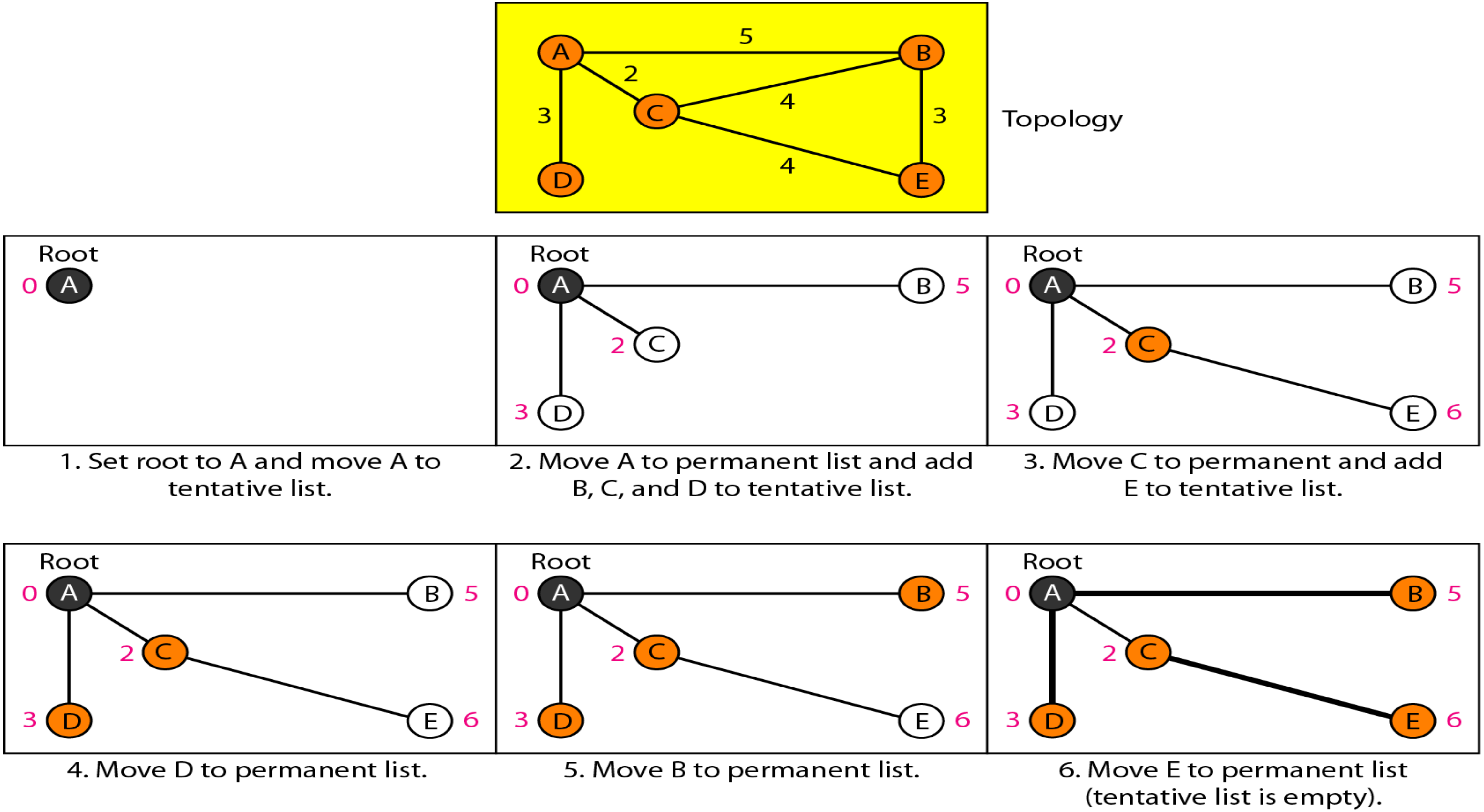
3. Formation of Shortest Path Tree: Dijkstra Algorithm

- A shortest path tree is a tree in which the path between the root and every other node is the shortest.
- The Dijkstra algorithm creates a shortest path tree from a graph. The algorithm divides the nodes into two sets: tentative and permanent.

It finds the neighbors of a current node, makes them tentative, examines them, and if they pass the criteria, makes them permanent.




Topology
A B C D E
5 2 4 3 3 4
Root
1. Set root to A and move A to tentative list.
2. Move A to permanent list and add B, C, and D to tentative list.
3. Move C to permanent and add E to tentative list.
4. Move D to permanent list.
5. Move B to permanent list.
6. Move E to permanent list (tentative list is empty).

4. Calculation of a routing table

Routing table for node A Path

Node	*Cost*	*Next Router*
A	0	—
B	5	—
C	2	—
D	3	—
E	6	C

OSPF

- Open Shortest Path First (OSPF) is a routing protocol for IP networks. It is used within a network or area. OSPF is an interior gateway protocol designed for a single autonomous system.
- OSPF uses a link-state routing algorithm. Each router has information about every link and router in the network. It finds the shortest path to each destination. OSPF learns about all routers and subnets in the network to build a link-state database (LSDB). Routers exchange link-state advertisements (LSAs) to share information about routers, subnets, and more.

Basic Terms and operations in OSPF

- Link-state : Description of a link between two routers
- SPF algorithm : Computes shortest path from a source router to others.
- OSPF cost : Metric representing the cost of using a link or path.
- Shortest path tree : Shows the shortest path from a source router to all others.
- Areas : Logical subdivisions of an OSPF network with similar characteristics.
- Border routers : Connect different areas or external networks.
- Link-state packets : Contain link-state advertisements and are sent by routers.

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OSPF Steps

- Neighbor Discovery Routers find and communicate with neighbors on the same link.
- Database Exchange, Routers exchange LSAs to learn about network topology.
- Route Calculation Routers use SPF algorithm to find the best paths.

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BGP

- BGP stands for Border Gateway Protocol. It is a standardized gateway protocol that exchanges routing information across autonomous systems (AS). When one network router is linked to other networks, it cannot decide which network is the best network to share its data to by itself.
- Border Gateway Protocol considers all peering partners that a router has and sends traffic to the router closest to the data destination. This communication is possible because, at boot, BGP allows peers to communicate their routing information and then stores that information in a Routing Information Base (RIB).
- The main goal of BGP is to find any path to the destination that is loop-free.
- The routers that connect other ASs are called border gateways. The task of the border gateways is to forward packets between ASs.

Types of Border Gateway Protocol

- External BGP: It is used to interchange routing information between the routers in different autonomous systems, it is also known as eBGP.
- Internal BGP: It is used to interchange routing information between the routers in the same autonomous system, it is also known as iBGP

Congestion Control Algorithm

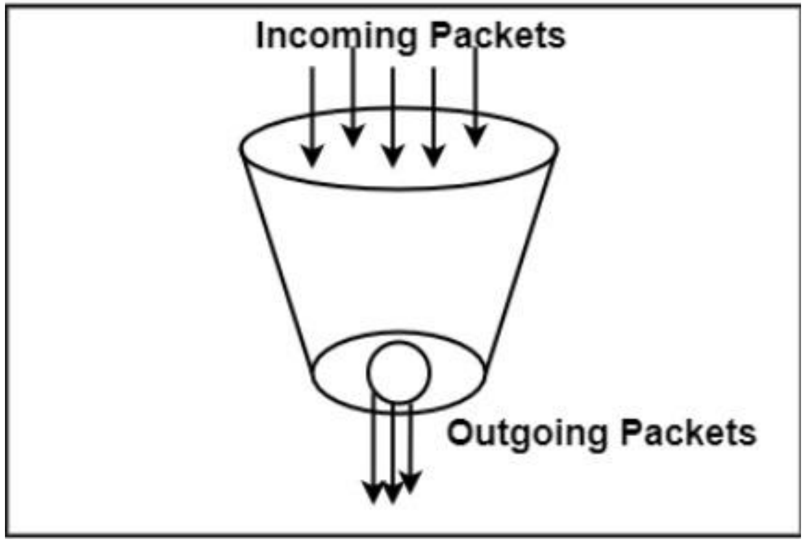
- Too many packets present in (a part of) the network causes packet delay and loss that degrades performance. This situation is called **congestion**. The network and transport layers share the responsibility for handling congestion
- There are two congestion control algorithms:

1. Leaky Bucket
2. Token Bucket Algorithm

Leaky Bucket

- The leaky bucket algorithm discovers its use in the context of network traffic shaping or rate-limiting. The algorithm allows controlling the rate at which a record is injected into a network and managing burstiness in the data rate.
- In this algorithm, a bucket with a volume of ‘b’ bytes and a hole in the bottom is considered. If the bucket is null, it means b bytes are available as storage. A packet with a size smaller than b bytes arrives at the bucket and will forward it. If the packet's size increases by more than b bytes, it will either be discarded or queued. It is also considered that the bucket leaks through the hole in its bottom at a constant rate of r bytes per second.
- The outflow is considered constant when there is any packet in the bucket and zero when it is empty. This defines that if data flows into the bucket faster than data flows out through the hole, the bucket overflows.
- The disadvantages compared with the leaky-bucket algorithm are the inefficient use of available network resources. The leak rate is a fixed parameter. In the case of the traffic, volume is deficient, the large area of network resources such as bandwidth is not being used effectively. The leaky-bucket algorithm does not allow individual flows to burst up to port speed to effectively consume network resources when there would not be resource contention in the network.

Leaky Bucket Algorithm



Token Bucket Algorithm

- The leaky bucket algorithm has a rigid output design at the average rate independent of the bursty traffic. In some applications, when large bursts arrive, the output is allowed to speed up. This calls for a more flexible algorithm, preferably one that never loses information. Therefore, a token bucket algorithm finds its uses in network traffic shaping or rate-limiting.
- It is a control algorithm that indicates when traffic should be sent. This order comes based on the display of tokens in the bucket. The bucket contains tokens. Each of the tokens defines a packet of predetermined size. Tokens in the bucket are deleted for the ability to share a packet.
- When tokens are shown, a flow to transmit traffic appears in the display of tokens. No token means no flow sends its packets. Hence, a flow transfers traffic up to its peak burst rate in good tokens in the bucket.
- Thus, the token bucket algorithm adds a token to the bucket each 1 / r seconds. The volume of the bucket is ‘b’ tokens. When a token appears, and the bucket is complete, the token is discarded. If a packet of n bytes appears and n tokens are deleted from the bucket, the packet is forwarded to the network.
- When a packet of n bytes appears but fewer than n tokens are available. No tokens are removed from the bucket in such a case, and the packet is considered non-conformant. The non-conformant packets can either be dropped or queued for subsequent transmission when sufficient tokens have accumulated in the bucket.
- They can also be transmitted but marked as being non-conformant. The possibility is that they may be dropped subsequently if the network is overloaded.

Quality-of-service (QoS)

- Quality-of-service (QoS) refers to traffic control mechanisms that seek to differentiate performance based on application or network-operator requirements or provide predictable or guaranteed performance to applications, sessions, or traffic aggregates. The basic phenomenon for QoS is in terms of packet delay and losses of various kinds.

QoS Parameters

- Packet loss: This occurs when network connections get congested, and routers and switches begin losing packets.
- Jitter: This is the result of network congestion, time drift, and routing changes.
- Latency: This is how long it takes a packet to travel from its source to its destination. The latency should be as near to zero as possible.
- Bandwidth: This is a network communications link's ability to transmit the majority of data from one place to another in a specific amount of time.
- Mean opinion score: This is a metric for rating voice quality that uses a five-point scale, with five representing the highest quality.

Quality of Service (QoS) ensures the performance of critical applications within limited network capacity.

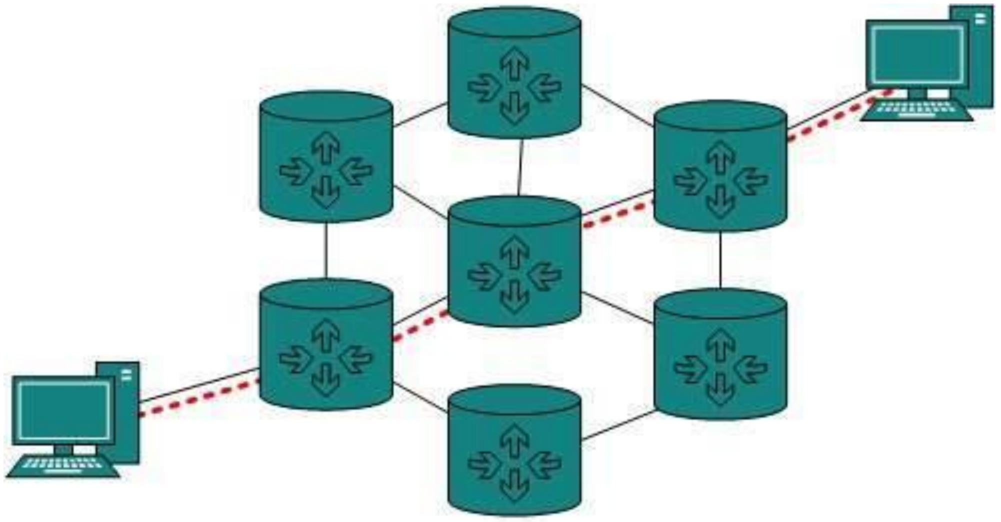
- Packet Marking: QoS marks packets to identify their service types.
- Virtual Queues: Routers create separate virtual queues for each application based on priority.
- Handling Allocation: QoS assigns the order in which packets are processed, ensuring appropriate bandwidth for each application

Benefits of QoS

- Improved Performance for Critical Applications
- Enhanced User Experience
- Efficient Bandwidth Utilization
- Increased Network Reliability
- Compliance with Service Level Agreements (SLAs)
- Reduced Network Costs
- Improved Security
- Better Scalability

Internetworking

- Routing between two networks is called internetworking or more simply an **internet**.
- Networks can be considered different based on various parameters such as, Protocol, topology, Layer-2 network and addressing scheme.
- In internetworking, routers have knowledge of each other's address and addresses beyond them. They can be statically configured on different network or they can learn by using internetworking routing protocol.
- Routing protocols which are used within an organization or administration are called Interior Gateway Protocols or IGP.
- Routing between different organizations or administrations may have Exterior Gateway Protocol, and there is only one EGP i.e. Border Gateway Protocol.



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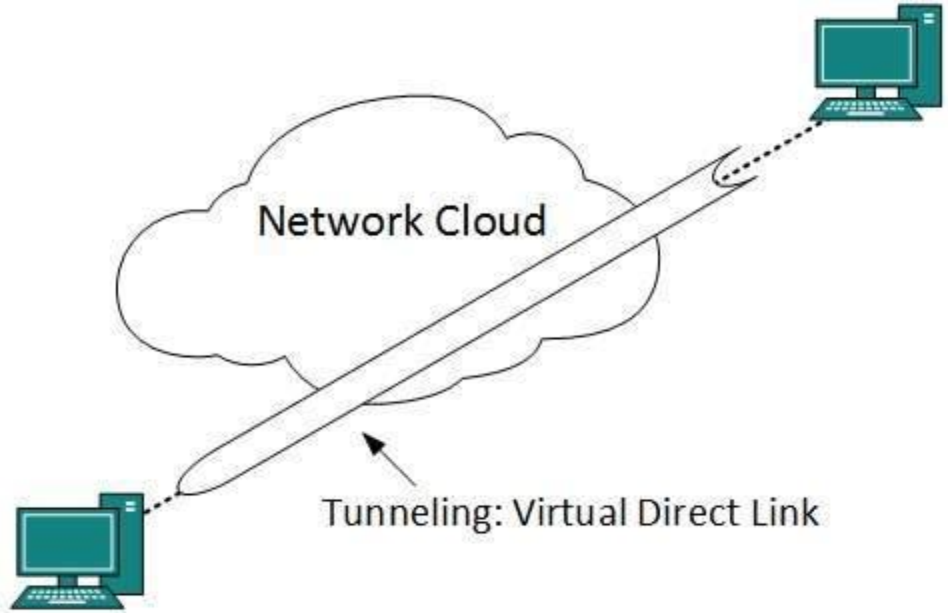
Tunneling

- If they are two geographically separate networks, which want to communicate with each other, they may deploy a dedicated line between or they have to pass their data through intermediate networks.
- Tunneling is a mechanism by which two or more same networks communicate with each other, by passing intermediate networking complexities. Tunneling is configured at both ends.



- When the data enters from one end of Tunnel, it is tagged. This tagged data is then routed inside the intermediate or transit network to reach the other end of Tunnel. When data exists the Tunnel its tag is removed and delivered to the other part of the network.
- Both ends seem as if they are directly connected and tagging makes data travel through transit network without any modifications.

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Packet Fragmentation

- Most Ethernet segments have their maximum transmission unit (MTU) fixed to 1500 bytes. A data packet can have more or less packet length depending upon the application. Devices in the transit path also have their hardware and software capabilities which tell what amount of data that device can handle and what size of packet it can process.
- If the data packet size is less than or equal to the size of packet the transit network can handle, it is processed neutrally. If the packet is larger, it is broken into smaller pieces and then forwarded. This is called packet fragmentation. Each fragment contains the same destination and source address and routed through transit path easily. At the receiving end it is assembled again.
- If a packet with DF (don't fragment) bit set to 1 comes to a router which can not handle the packet because of its length, the packet is dropped.
- When a packet is received by a router has its MF (more fragments) bit set to 1, the router then knows that it is a fragmented packet and parts of the original packet is on the way.
- If packet is fragmented too small, the overhead is increases. If the packet is fragmented too large, intermediate router may not be able to process it and it might get dropped.

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The Network Layer in the Internet

- There are some principles that drove its design in the past and made it the success that it is today. They are:

1. **Make sure it works.** Do not finalize the design or standard until multiple prototypes have successfully communicated with each other.
2. **Keep it simple.** When in doubt, use the simplest solution.
3. **Make clear choices.** If there are several ways of doing the same thing, choose only one.
4. **Exploit modularity.** This principle leads directly to the idea of having protocol stacks, each of whose layers is independent of all the other ones.
5. **Expect heterogeneity.** Different types of hardware, transmission facilities, and applications will occur on any large network. To handle them, the network design must be simple, general, and flexible.
6. **Avoid static options and parameters.**
7. **Look for a good design; it need not be perfect.**
8. **Be strict when sending and tolerant when receiving.**
9. **Think about scalability.**
10. **Consider performance and cost.**

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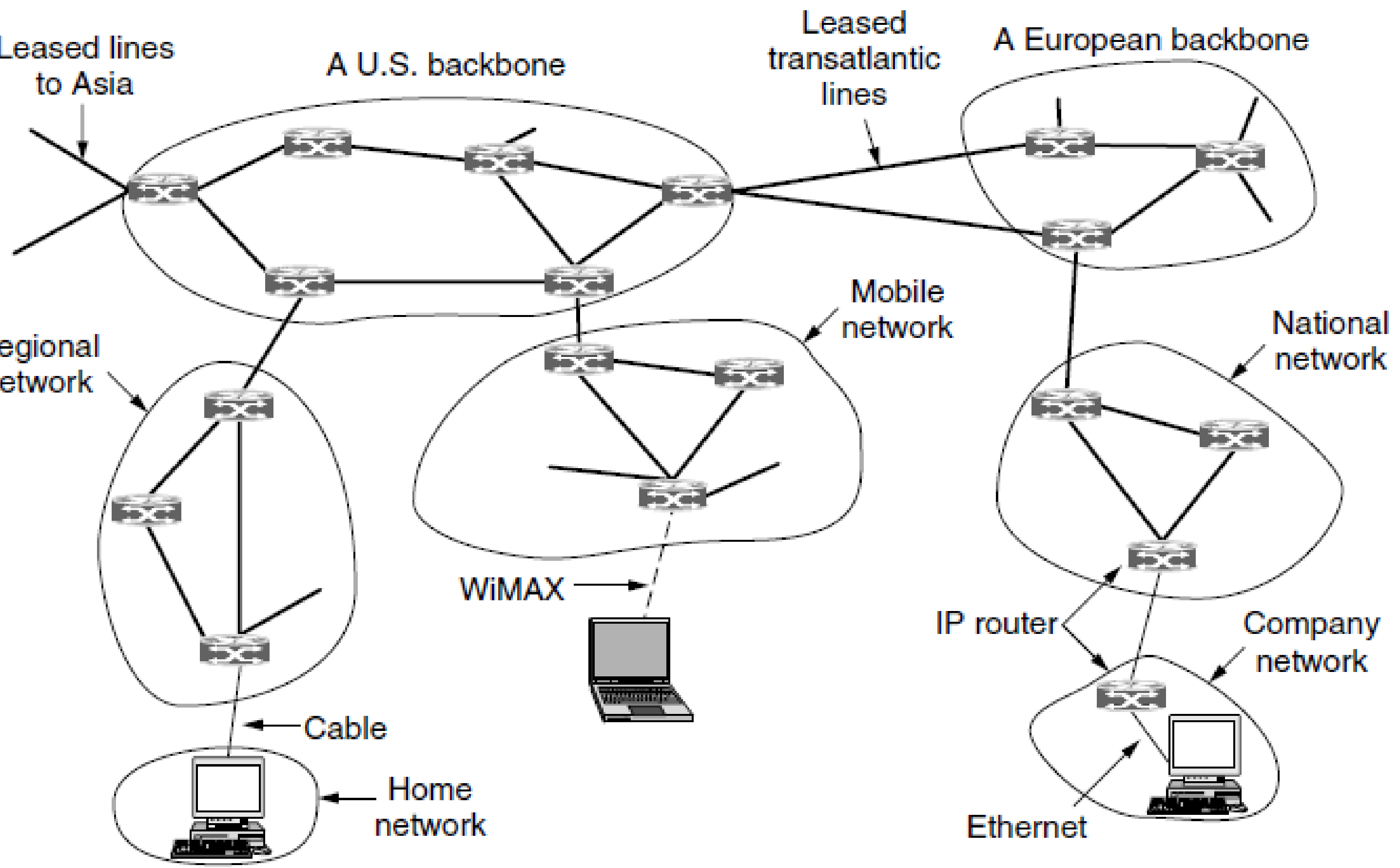
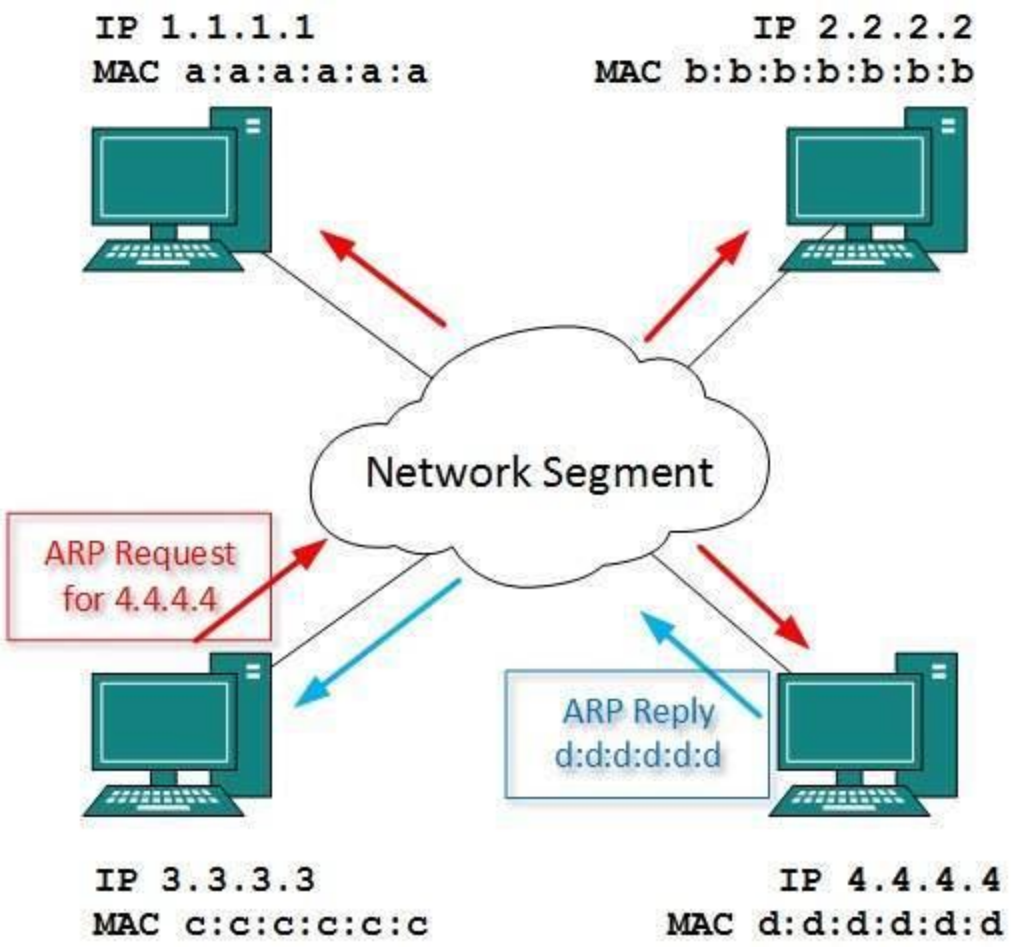


The Internet is an interconnected collection of many networks.

Every computer in a network has an IP address by which it can be uniquely identified and addressed. An IP address is Layer-3 (Network Layer) logical address. This address may change every time a computer restarts. A computer can have one IP at one instance of time and another IP at some different time.

Address Resolution Protocol(ARP)

- While communicating, a host needs Layer-2 (MAC) address of the destination machine which belongs to the same broadcast domain or network. A MAC address is physically burnt into the Network Interface Card (NIC) of a machine and it never changes.
- On the other hand, IP address on the public domain is rarely changed. If the NIC is changed in case of some fault, the MAC address also changes. This way, for Layer-2 communication to take place, a mapping between the two is required.
- To know the MAC address of remote host on a broadcast domain, a computer wishing to initiate communication sends out an ARP broadcast message asking, “Who has this IP address?” Because it is a broadcast, all hosts on the network segment (broadcast domain) receive this packet and process it. ARP packet contains the IP address of destination host, the sending host wishes to talk to. When a host receives an ARP packet destined to it, it replies back with its own MAC address.



Once the host gets destination MAC address, it can communicate with remote host using Layer-2 link protocol. This MAC to IP mapping is saved into ARP cache of both sending and receiving hosts. Next time, if they require to communicate, they can directly refer to their respective ARP cache.

Reverse ARP is a mechanism where host knows the MAC address of remote host but requires to know IP address to communicate.

Internet Control Message Protocol (ICMP)

- ICMP is network diagnostic and error reporting protocol. ICMP belongs to IP protocol suite and uses IP as carrier protocol. After constructing ICMP packet, it is encapsulated in IP packet. Because IP itself is a best-effort non-reliable protocol, so is ICMP.
- Any feedback about network is sent back to the originating host. If some error in the network occurs, it is reported by means of ICMP. ICMP contains dozens of diagnostic and error reporting messages.
- ICMP-echo and ICMP-echo-reply are the most commonly used ICMP messages to check the reachability of end-to-end hosts. When a host receives an ICMP-echo request, it is bound to send back an ICMP-echo-reply. If there is any problem in the transit network, the ICMP will report that problem.

Internet Protocol Version 4 (IPv4)

- IPv4 is 32-bit addressing scheme used as TCP/IP host addressing mechanism. IP addressing enables every host on the TCP/IP network to be uniquely identifiable.
- IPv4 provides hierarchical addressing scheme which enables it to divide the network into sub-networks, each with well-defined number of hosts. IP addresses are divided into many categories:
- **Class A** - it uses first octet for network addresses and last three octets for host addressing
- **Class B** - it uses first two octets for network addresses and last two for host addressing
- **Class C** - it uses first three octets for network addresses and last one for host addressing
- **Class D** - it provides flat IP addressing scheme in contrast to hierarchical structure for above three.
- **Class E** - It is used as experimental.
- IPv4 also has well-defined address spaces to be used as private addresses (not routable on internet), and public addresses (provided by ISPs and are routable on internet).
- Though IP is not reliable one; it provides 'Best-Effort-Delivery' mechanism.

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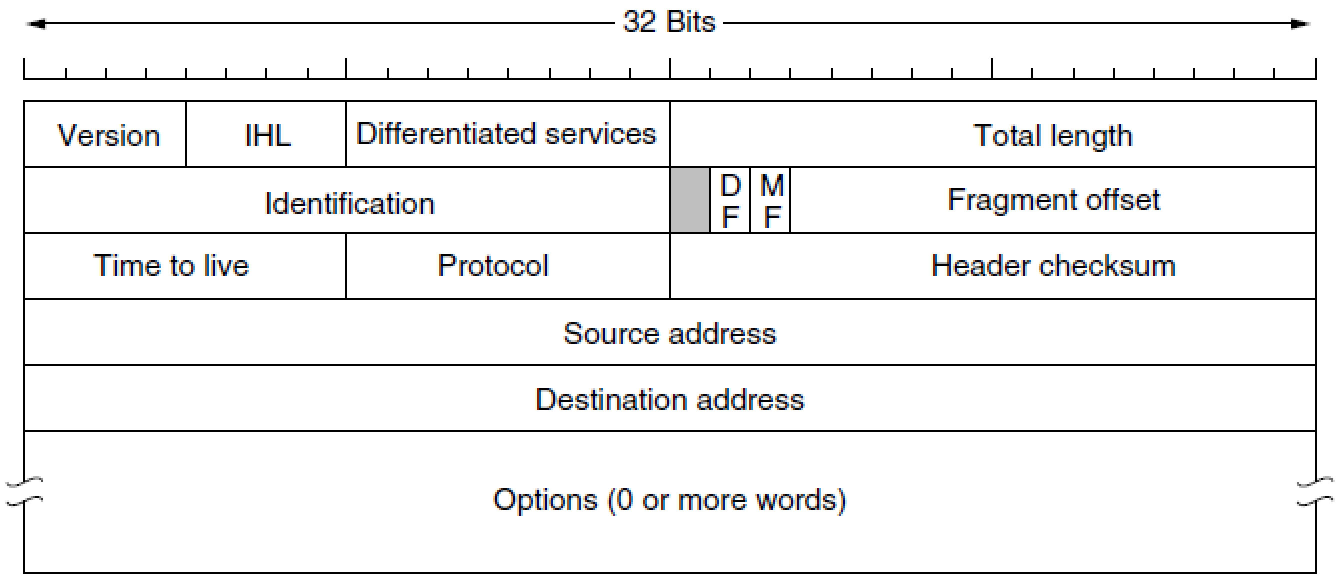
The IPv4 (Internet Protocol) header.



- The *Version* field keeps track of which version of the protocol the datagram belongs to.
- *IHL*, (Internet Header Length)is provided to tell how long the header is, in 32-bit words.
- The *Differentiated services* field specifies *Type of service*
- The *Total length* includes everything in the datagram—both header and data. The maximum length is 65,535 bytes.
- The *Identification* field is needed to allow the destination host to determine which packet a newly arrived fragment belongs to.
- Then come two 1-bit fields related to fragmentation. *DF* stands for Don't Fragment. *MF* stands for More Fragments.
- The *Fragment offset* tells where in the current packet this fragment belongs.

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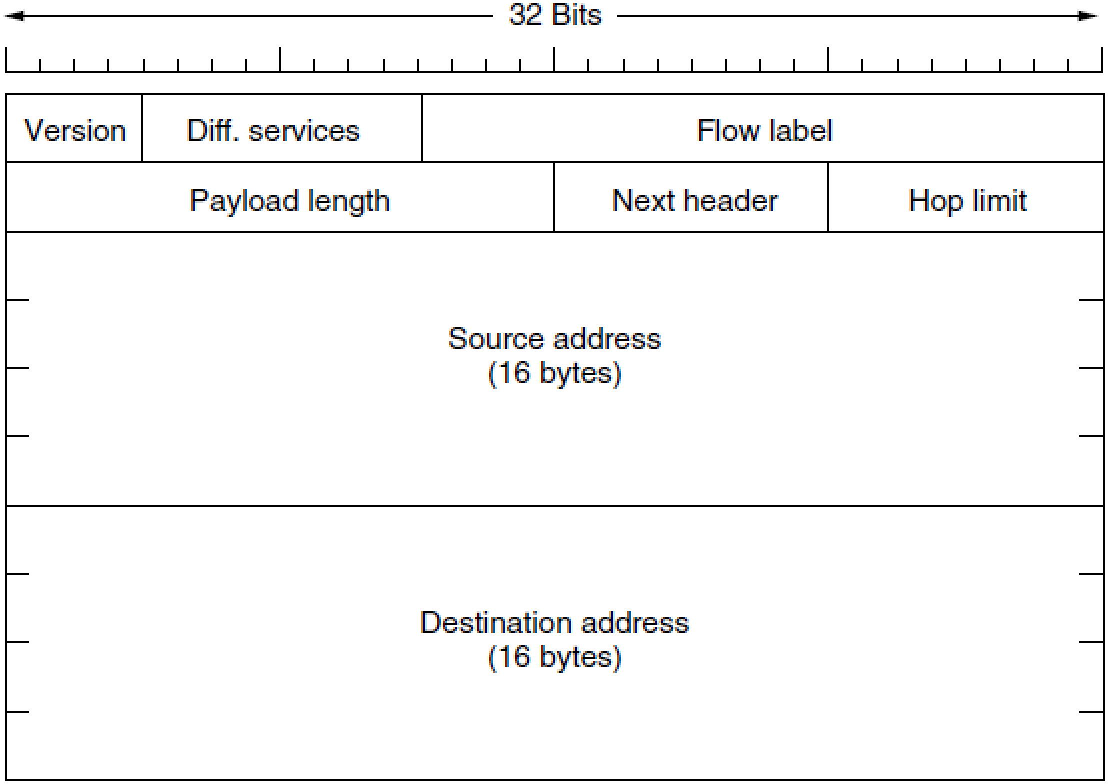
- The *TtL (Time to live)* field is a counter used to limit packet lifetimes.
- The *Protocol* field tells it which transport process to give the packet to.
- *Header checksum:* Since the header carries vital information such as addresses, it rates its own checksum for protection.
- The *Source address* and *Destination address* indicate the IP address of the source and destination network interfaces.
- The *Options* field was designed to provide an escape to allow subsequent versions of the protocol to include information not present in the original design.

Internet Protocol Version 6 (IPv6)

- Exhaustion of IPv4 addresses gave birth to a next generation Internet Protocol version 6. IPv6 addresses its nodes with 128-bit wide address providing plenty of address space for future to be used on entire planet or beyond.
- IPv6 has introduced Anycast addressing but has removed the concept of broadcasting. IPv6 enables devices to self-acquire an IPv6 address and communicate within that subnet. This auto-configuration removes the dependability of Dynamic Host Configuration Protocol (DHCP) servers. This way, even if the DHCP server on that subnet is down, the hosts can communicate with each other.
  - Dynamic Host Configuration Protocol (DHCP) is a network management protocol used to dynamically assign an IP address to any device, or node, on a network so they can communicate using IP (Internet Protocol). There is no need to manually assign IP addresses to new devices.
- IPv6 provides new feature of IPv6 mobility. Mobile IPv6 equipped machines can roam around without the need of changing their IP addresses.
- IPv6 is still in transition phase and is expected to replace IPv4 completely in coming years.

The IPv6 fixed header (required).



- The *Flow label* field provides a way for a source and destination to mark groups of packets that have the same requirements and should be treated in the same way by the network
- The *Payload length* field tells how many bytes follow the byte header
- The *Next header* field tells which transport protocol handler (e.g., TCP, UDP) to pass the packet to.
- The *Hop limit* field is same as the *Time to live* field in IPv4

Dynamic Host Configuration Protocol (DHCP)

- Dynamic Host Configuration Protocol is a network protocol used to automate the process of assigning IP addresses and other network configuration parameters to devices (such as computers, smartphones and printers) on a network.
- Instead of manually configuring each device with an IP address, DHCP allows devices to connect to a network and receive all necessary network information, like IP address, subnet mask, default gateway and DNS server addresses, automatically from a DHCP server.

To make sure your DHCP servers are safe, consider these DHCP security issues:

- ***Limited IP Addresses*** : A DHCP server can only offer a set number of IP addresses. This means attackers could flood the server with requests, causing essential devices to lose their connection.
- ***Fake DHCP Servers*** : Attackers might set up fake DHCP servers to give out fake IP addresses to devices on your network.
- ***DNS Access*** : When users get an IP address from DHCP, they also get DNS server details. This could potentially allow them to access more data than they should.

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Protection Against DHCP Starvation Attack

- A DHCP starvation attack happens when a hacker floods a DHCP server with requests for IP addresses. This overwhelms the server, making it unable to assign addresses to legitimate users. The hacker can then block access for authorized users and potentially set up a fake DHCP server to intercept and manipulate network traffic, which could lead to a man-in-the-middle attack.

Network Address Translation (NAT)

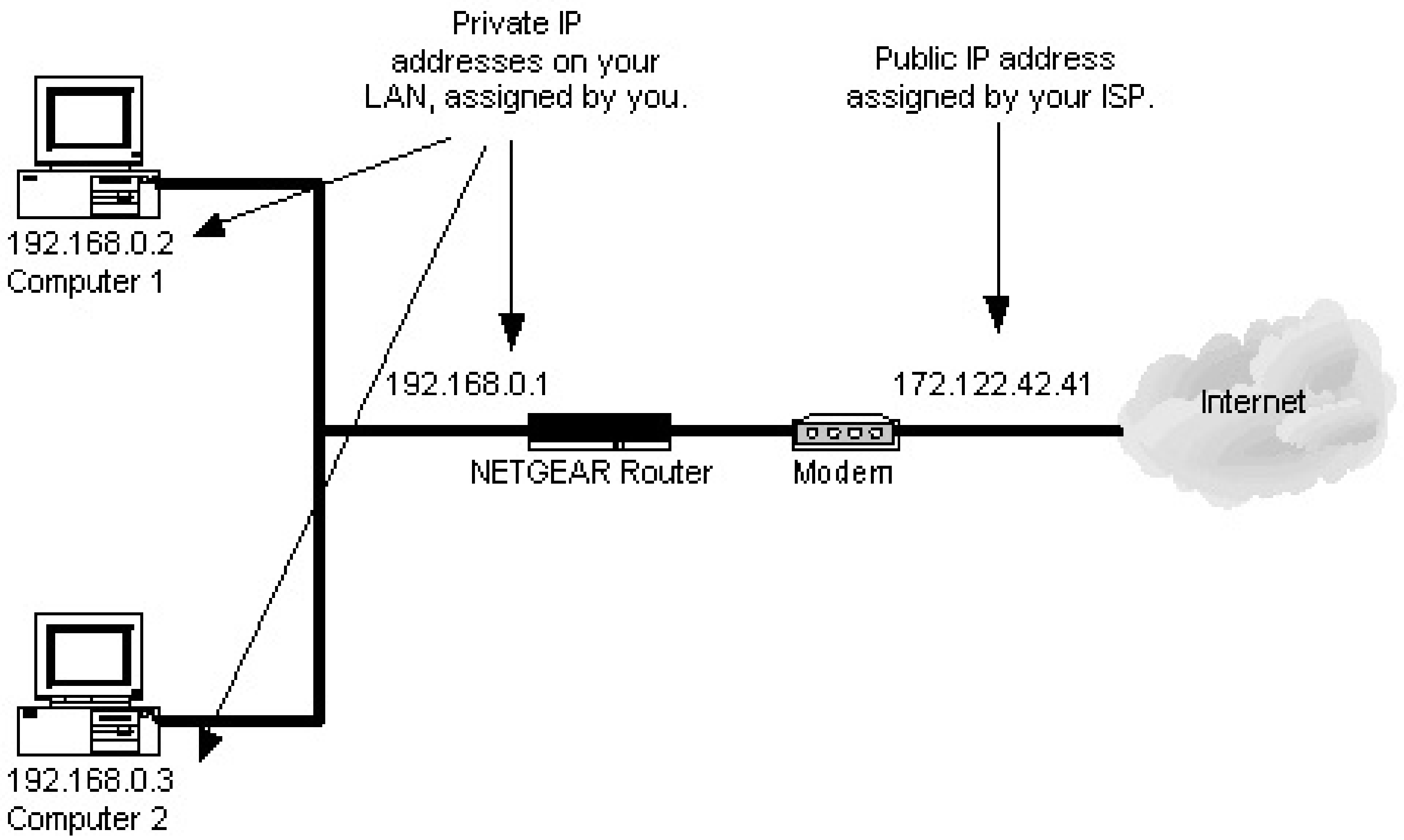
We generally have two types of IP address, which are as follows

- Private IP address : used in the LAN
- Public IP address: provided by the ISP is configured in the WAN

Public IP addresses are always paid, while the private IP address is free.

- Network Address translation is used to translate Private IP address to Public IP address to communicate LAN side of the Device to Global Network. Network address translation can be processed in Router or Firewall.

- Usually we use gateway router / Border devices for NAT configuration. One of the interfaces for that device is connected to the local Area network (INSIDE) and one of the interfaces for this device connected to the outside network (OUTSIDE).
- When there is a request from local machine it will hit the configuration pool then that Private IP will convert it into Public IP address and vice versa.



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That's all about

UNIT 3

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